Flying Blind: Comparing SIFT and ORB for GPS-Free Drone Localization under Image Degradation

Digital Image Processing – Spring 2026

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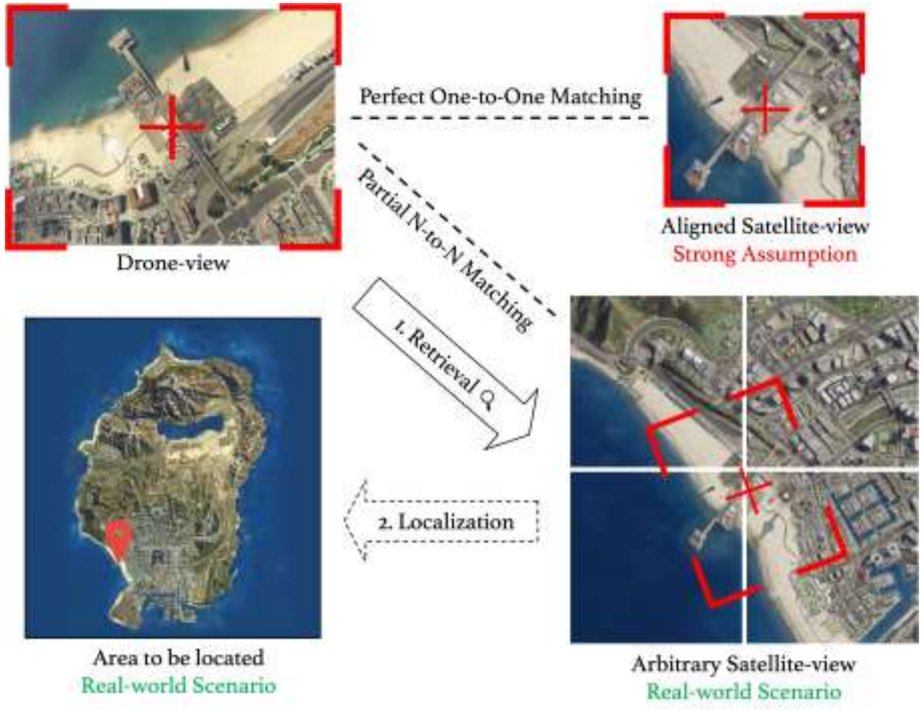
Motivation

- GPS can be unavailable or degraded in many environments (urban areas, signal interference)

- Drones need accurate localization: safe navigation depends on knowing the drone's position

- Alternative: visual localization
  - Use onboard camera instead of GPS
  - Match drone images to satellite maps
  - Estimate position from visual correspondence

- Real-world conditions are challenging
  - Motion blur from fast movement
  - Noise from sensors
  - Low-resolution imagery
  - Satellite/drone viewpoint differences

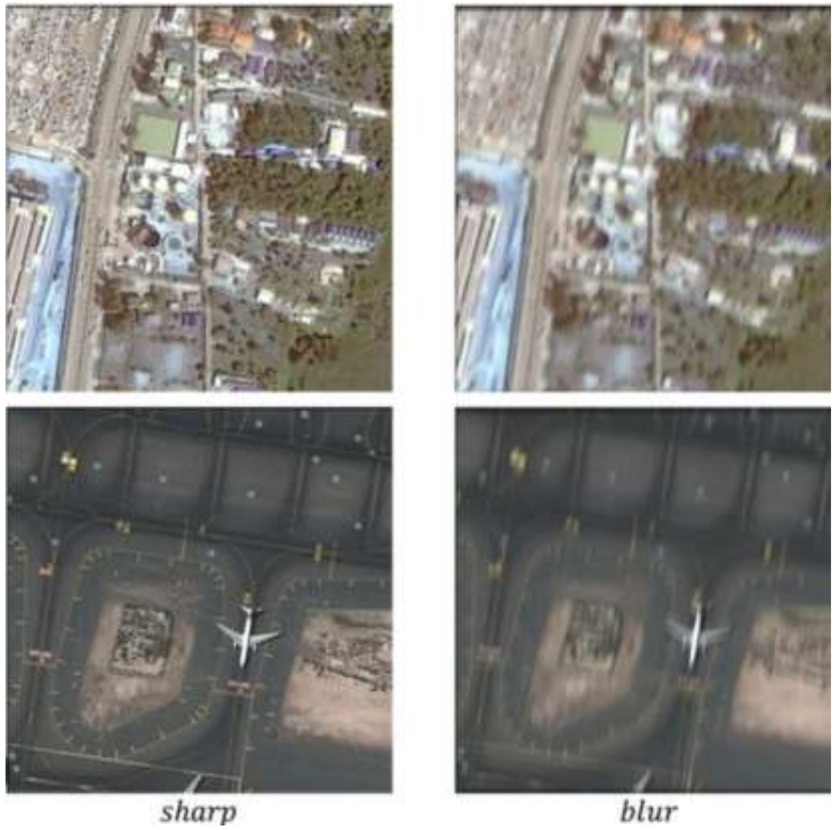


Prior Work & Limitations

- SIFT (Scale Invariant Feature Transform)
  - Robust feature-based method widely used for cross-view image matching
  - Strong invariance to scale and rotation, but computationally expensive

- ORB (Oriented FAST and Rotated BRIEF)
  - Fast and efficient alternative to SIFT for real-time applications
  - More sensitive to large viewpoint and scale changes

- Gap in prior work
  - Most studies assume near-ideal image conditions (clean, high-resolution data)
  - Limited analysis of performance under real world degradations (blur, noise, resolution loss)



Dataset

- High geographic and visual diversity, but experiments focus on a small controlled subset for feasibility

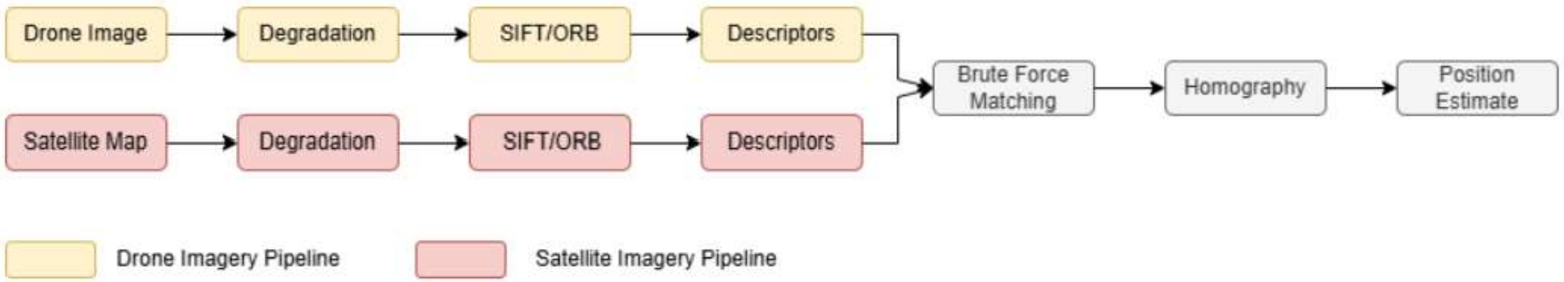
- UAV-VisLoc dataset
  - 6,742 drone images + 11 satellite maps across 11 locations in China
  - Includes ground-truth GPS for evaluation

- Study subset (for computational efficiency)
  - Single location: Taizhou-1 (China) 1 satellite map + 768 drone images
  - Coverage: ~8.86 km × 7.25 km (≈64 km² area)
  - Used subset of 5 drone images for experiments (13 hours to run)

Pipeline Overview

- Experimental setup
  - A baseline using clean, unmodified images
  - Each degradation type is applied independently to: drone images/ satellite maps
  - isolates the effect of each degradation on localization performance

- Pipeline description below:



Degradation Experiments

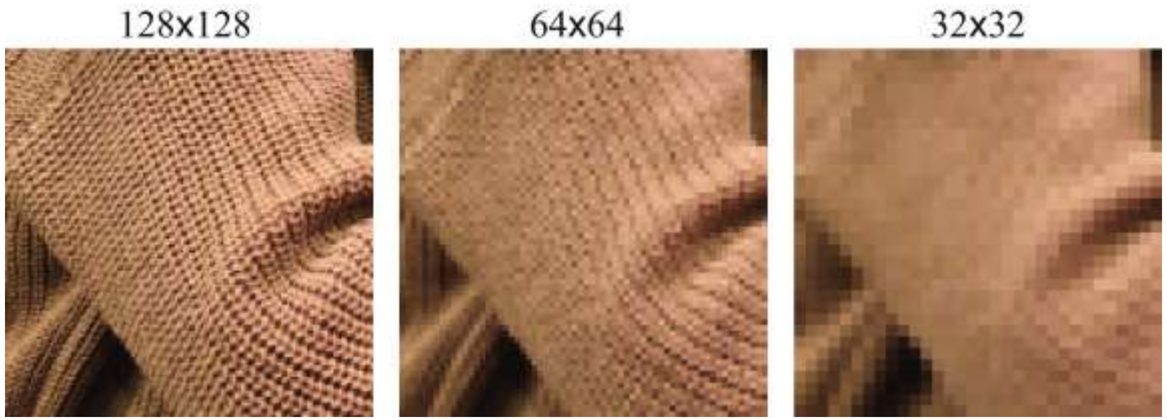
Drone Image Degradation

Degradation	Method	Parameters
Motion Blur	Linear Convolution Kernel	Kernel size: 5, 15
Gaussian Noise	Additive Noise	Sigma: 10, 30
Resolution Reduction	Downsample	Factor: 2, 4

Satellite Image Degradation

Degradation	Method	Parameters
Resolution Reduction	Downsample	Factor: 2, 4

- One degradation type applied at a time
- Both SIFT and ORB evaluated under each condition



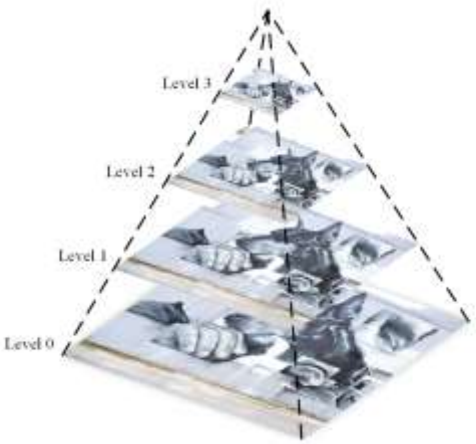
Feature Detection

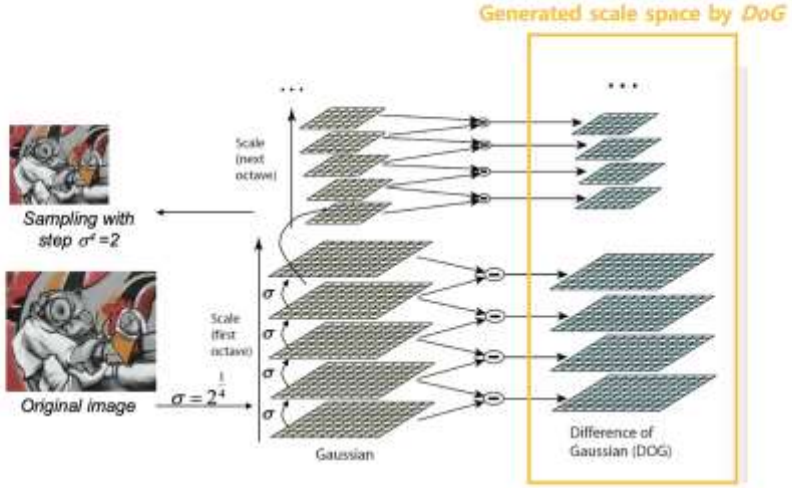
- Similarities:
  - Both detect local keypoints and compute descriptors for matching
  - Both use a scale pyramid to achieve scale awareness

- Differences:

	SIFT	ORB
Keypoint detection	Difference of Gaussians	FAST corner detector
Descriptor	128 float vector	256-bit binary string
Distance Metric	L2	Hamming
Scale invariance	Strong (multi scale representation)	Moderate (limited pyramid)
Speed	Slow	100x faster

- SIFT trades speed for accuracy and scale robustness
- ORB trades accuracy for speed and noise robustness

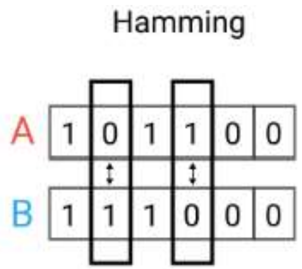


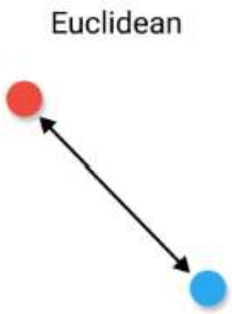


Matching & Position Estimation

Descriptor Matching

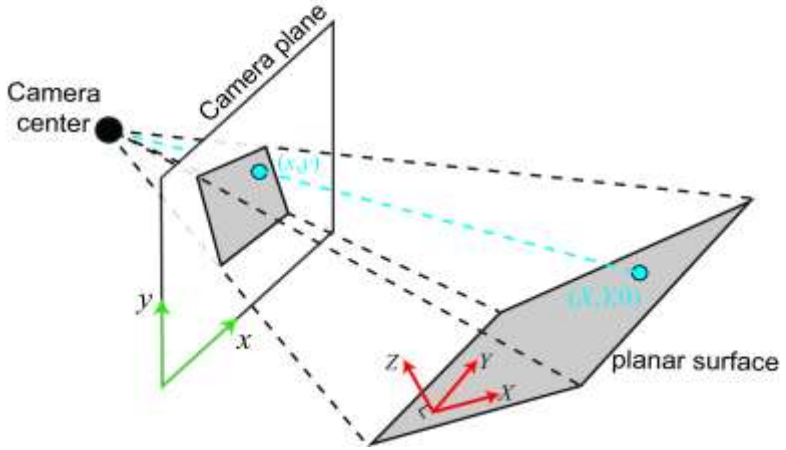
- Satellite map is divided into overlapping tiles
- Each tile is matched against the drone image
  SIFT: float descriptors compared with L2 distance
- ORB: binary descriptors compared with Hamming distance
- Lowe's ratio test: only keep match if the best match is significantly better than the other





Position Estimation

- RANSAC computes a homography from the filtered matches (Lowe’s ratio test)
- Homography maps the drone image to the satellite image
- Deduce GPS location from the transform

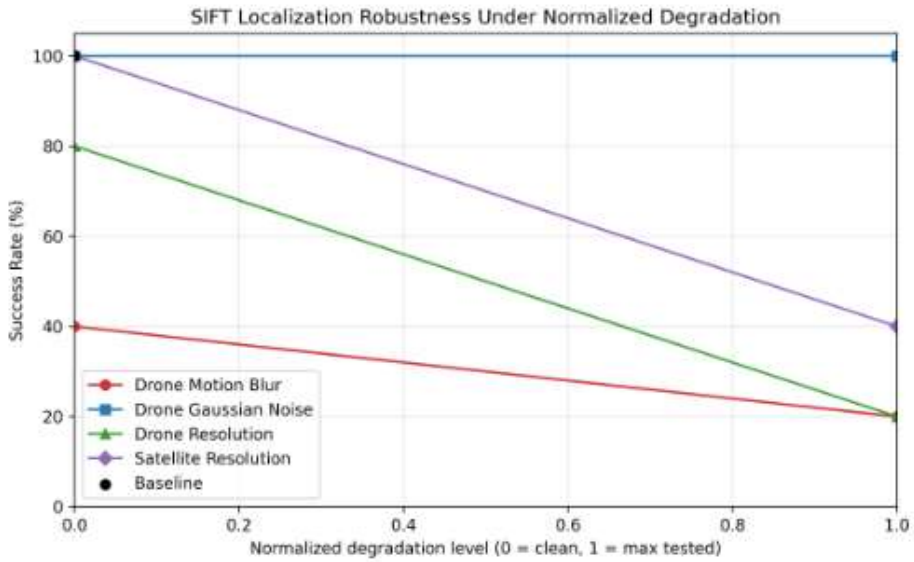


Results

- ORB failed entirely across all experiments including the clean baseline: not suitable for cross-view UAV to satellite matching
- SIFT achieved 100% success on clean images but with large localization errors
- Motion blur is the most damaging degradation rate drops to 40% at kernel 5 and 20% at kernel 15

TABLE I
LOCALIZATION PERFORMANCE UNDER ALL EXPERIMENTAL CONDITIONS

Experiment	Detector	Success Rate (%)	Mean Error (m)	Median Error (m)
Baseline	SIFT	100	4785.32	3493.15
	ORB	0	–	–
Drone Motion Blur — kernel 5	SIFT	40	2270.25	2270.25
	ORB	0	–	–
Drone Motion Blur — kernel 15	SIFT	20	4678.31	4678.31
	ORB	0	–	–
Drone Gaussian Noise — $\sigma = 10$	SIFT	100	3962.81	3493.15
	ORB	0	–	–
Drone Gaussian Noise — $\sigma = 30$	SIFT	100	5543.48	6427.96
	ORB	0	–	–
Drone Resolution — factor 2	SIFT	80	4341.39	4341.39
	ORB	0	–	–
Drone Resolution — factor 4	SIFT	20	5079.51	5079.51
	ORB	0	–	–
Satellite Resolution — factor 2	SIFT	100	4681.72	4681.72
	ORB	0	–	–
Satellite Resolution — factor 4	SIFT	40	4098.36	4098.36
	ORB	0	–	–



- Gaussian noise has little effect SIFT: 100% success at both noise levels
- Resolution reduction: damaging for both drone and satellite imagery but satellite degradation is less severe
- All results based on 5 images trends are indicative but should be interpreted with caution

Conclusion

Findings

- ORB failed entirely across all conditions
- SIFT remained robust across most conditions
- Motion blur is the most damaging degradation
- Gaussian noise has little effect on SIFT
- Satellite degradation is less damaging than drone image degradation

Limitations & Future Work

- 5 images tested: not statistically conclusive
- Test on larger image subsets
- Investigate preprocessing to improve cross-view matching
- Explore tuning ORB's scale pyramid for high-altitude imagery
- Consider hybrid or adaptive approaches